BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 232 · OCR pseudocode, files and IDEs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. If y=4 is passed by ref and a method adds 5, what is y after the call?

Q2. What does a breakpoint do?

Q3. Which condition controls a while loop?

Q4. If a file has two lines and each readLine reads one line, how many calls read the whole file?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

openRead — Open a file for reading using OCR A Level pseudocode.

readLine — Read the next line from an open file.

endOfFile — Test whether no unread lines remain.

openWrite — Open for writing; existing file contents are replaced.

close — Release an opened file resource.

New learning

Open the resource — openRead returns a file object for reading; it does not overwrite the file.

Read until EOF — endOfFile() is false while another line remains. readLine() retrieves one line and moves the file position.

Close the file — After the final line, endOfFile() is true, so the loop stops and the file is closed. For writing use openWrite , writeLine , then close() ; unlike reading, opening for writing overwrites any existing file contents .

Worked example

Trace OCR pseudocode that reads the first two lines of fruits.txt.

1. Predict first — The first call returns Apple and advances the file position.

2. Model and check — The next call returns Pear; variable a stays Apple.

3. Model and check — close releases the file resource; Plum has not been read.

4. Write mode is not read mode — openWrite opens the named text file for writing, replacing any previous contents. Do not use it simply to inspect scores already stored there.

5. Two calls create two lines — After the first call the file contains Ada,82 . After the second it also contains Ben,74 on the next line. A text file is not automatically a database table.

6. Predict a later read — f.close() releases the file. A separate later openRead("scores.txt") would read Ada,82 then Ben,74 in that order.

We Do · Practise with guidance

Trace a loop reading all lines of cities.txt. How many readLine calls occur?

```
f=openRead("cities.txt")
while NOT f.endOfFile()
  print(f.readLine())
endwhile
f.close()
```

Predict first

The loop prints York from the first readLine call.

Model and check

Then it prints Bath from the second call. At EOF the loop condition becomes false, so it stops after two reads.

Write two event messages to events.txt, then explain whether an old line survives.

```
log = openWrite("events.txt")
log.writeLine("START")
log.writeLine("DONE")
log.close()
```

List the final file lines

Predict the order of the two lines stored by the code and name the method that releases the file.

Two lines, then close

The file contains START and DONE on separate lines. close() finishes and releases the file.

Overwriting is deliberate

If events.txt previously contained OLD , it is not retained: this openWrite example overwrites rather than appends. To read existing lines use openRead .

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

An OCR pseudocode file contains Cat, Dog, Fox on separate lines. What do the first two readLine() calls return?

Question 2

Write OCR A Level pseudocode to open words.txt, display each line and close the file.

Question 3

Explain why openWrite is unsuitable for reading an existing file. Write OCR pseudocode that replaces events.txt with one line HELLO, and identify a debugging step if a reading loop omits its last record.

Exit Ticket · Answer before checking

OCR's openWrite("data.txt") safely reads existing data without altering it.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 9.

Q2. Pauses execution at a specified line.

Q3. A Boolean test evaluated before each iteration.

Q4. Two calls, assuming both lines are read and no additional data.

You Do pages · Independent questions

Q1. [4] First returns Cat [2]; second returns Dog [2]. Fox remains unread.

Q2. [6] f=openRead("words.txt") [1]; while NOT f.endOfFile() [2]; print(f.readLine()) [2]; endwhile, f.close() [1].

Q3. [8] openWrite replaces contents rather than reading [2]; use openRead for existing data [1]; f=openWrite("events.txt") [1], f.writeLine("HELLO") [1], f.close() [1]; breakpoint/step through final loop iteration [1]; inspect EOF condition/readLine position [1].

Exit Ticket · Model answer

False. openWrite opens the file for writing and replaces existing content.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.